Spatial Analysis of Private Health Infrastructure - Gurugram

Major Project Part 1

Submitted by

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In partial fulfillment of the requirement for the

Degree of Master of Technology in

URBAN DEVELOPMENT and MANAGEMENT

Submitted to

Department of Energy and Environment

TERI School of Advanced Studies

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DECLARATION

This is to certify that the work that forms the basis of this project **Spatial Analysis of Private Health Infrastructure - Gurugram** is an original work carried out by me and has not been submitted anywhere else for the award of any degree. I certify that all sources of information and data are fully acknowledged in the project report.

Signature

S. Manohar.

Sattaru Manohar

Gurugram
29-08-19

Place and Date

CERTIFICATE

This is to certify that **Sattaru Manohar** has carried out his major project in partial fulfilment of the requirement for the degree of Master of Technology in Urban Development and Management on the topic **Spatial Analysis of Private Health Infrastructure-Gurugram** during June 2019 to August 2019. The project was carried out at the **Gurugram Metropolitan Development Authority.**

The report embodies the original work of the candidate to the best of our knowledge.

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ABSTRACT

Indian cities are experiencing a rapid growth and most of the population are migrating to urban areas for their livelihood and putting pressure on social infrastructure.one of the main aspect of social infrastructure is Health infrastructure. Especially Healthcare sector plays a very crucial role in the enhancement of the citizens and also improves the economical standards of the citizens and helps in sustainable development of communities. So, it is necessary to focus on delivering the quality of services in urban areas and also very difficult to plan social infrastructure planning with conventional methods because it is highly cost intensive. Now a days it become easy with the advancement of technology and availability of GIS based spatial analysis tools it is very easy to plan and produce dynamic maps. For the purpose of spatial analysis of private health infrastructure in the research work and processing data sets in ARCMAP to produce map for the existing private health infrastructure in the study area Gurugram. Data gathered from GIS division and buffer zones are created for clinics around 1 km, Hospitals around 2km, Multispecialty around 5 km and Super specialty around 10km. Accessibility to existing Private health infrastructure also analyzed with 100- and 500-meters buffer zones along the road networks to check whether the health infrastructure is accessible from the major road networks . Along with that a 2031 projected population density of sectors also calculated to draw a relation weather the existing healthcare facility have the capacity to cater the projected data and also with the standards according to the URDPFI guidelines. In conclusion analyzing the data in ARCMAP can find the lack of private health infrastructure spatially and at the same can also check the necessity of the healthcare facilities for the growing population.

Key words: Healthcare, Infrastructure, GIS, Spatial Analysis, Gurugram.

Chapter 1

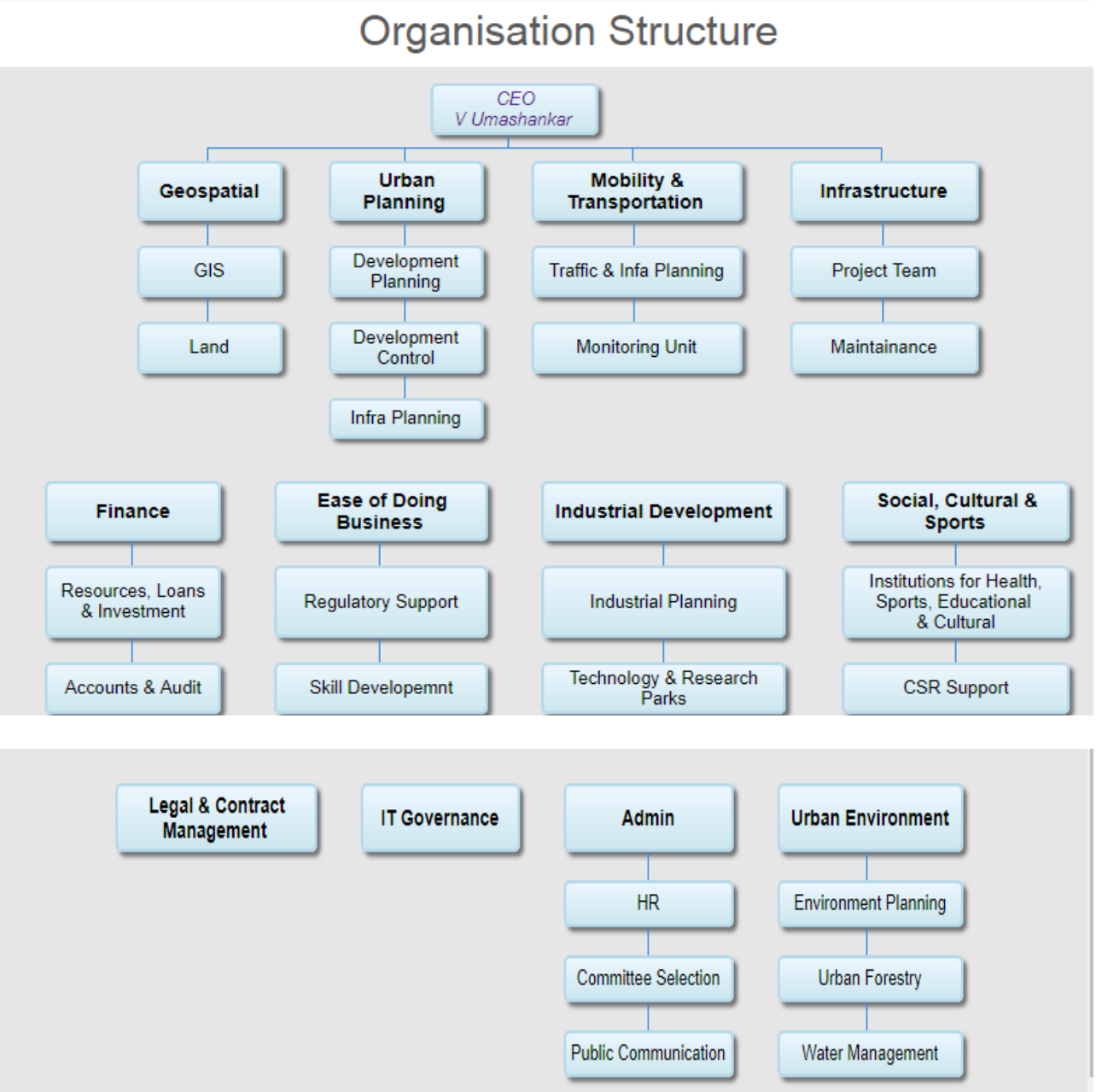
INTRODUCTION TO GURUGRAM METROPOLITAN DEVELOPMENT AUTHORITY (GMDA):

Initially Haryana Urban Development Authority taken care of planning and approvals for Gurugram area. But is the last few years the development activities increased at a faster rate due to job opportunities and sudden raise in real estate and construction sector. The population growth rate is increased very rapidly with the migrant population. Lot of unapproved constructions are started near villages because and land prices are sky rocketed every year and private players are taken huge parcels of lands from farmers and which results in fragmented development with improper road connectivity. These all leading to new environment and resources challenges in the Gurugram area and also degrading the quality and standard of life to the citizens. Due to improper connectivity the travelling cost to access the internal areas of sectors is getting increased and no regulation to private transporters like autos and the capacity of public transport is poor with the floating population. The pressure on natural resources like water, sewerage connections and huge waste generation leading to decrement of urban environment and polluting ground water and quality of air. In today scenario Gurugram emerged as the one of Information technology Hub and industrial hub in India and contributing 3rd highest per capital income to government of India and 48 to 50% of revenue to the government of Haryana. So, it is very difficult to manage the approval of building plans with limited resources.

To cope up with the emerging challenges Haryana government established a special authority to stream line the construction approvals and also plan the metropolitan area to cater the increasing population. GMDA was established under THE GURUGRAM METROPOLITAN DEVELOPMENT AUTHORITY ACT, 2017 (HARYANA ACT NO. 34 OF 2017) by the Government of Haryana with a vision to develop Gurugram Metropolitan Area in a sustainable and equitable growth by providing good quality of life and affordable standard of living to citizens through job creation opportunities, to provide for inclusive and integrated planning, infrastructure enhancement and provision of basic urban amenities and services, city level mobility management, sustainable management of social, economic, industrial development and urban environment, to reorganize the institutional structure of urban governance in coordination with local authorities in the situation of the emergence of Gurugram as a rapidly increasing urban agglomeration.

Organization Structure of GMDA:

Gurugram metropolitan development authority consists of 12 divisions and each department is assigned with particular development, enforcement and monitoring works.



Source: GMDA website.

Figure 1: GMDA Organization structure.

Chapter 2

URBAN PLANNING DIVISION:

The Urban Planning division main work is town planning. Development authority also responsible to prepare the Master Development Plan for Gurugram metropolitan area and also acts as the Major regulatory agency for the following approvals:

1. Development controls grant and change of land use permission.
2. Approval and Assessment of building plans
3. Approval of Building occupancy certificate
4. Extension of change of Land use
5. Revalidation of building plan
6. Stadium booking
7. New water connection
8. New sewerage, storm water drainage, and recycle water connection
9. Access permission to sector road for fuel station
10. water tanker and traffic light complaints

Citizens can access the above services through online service portal i.e.www.gmda.gov. in. The delivery time depends on the type of service they want and almost it takes 1 week to 1 months of time with nominal charges for the approvals.

The hierarchy of the urban planning department is shown in the below flow chart.

Source: Extracted by Author from GMDA ,2019.

Figure 2: Urban Planning division structure

Basically, building plan approvals will take long time duration for approval because it will go through a long process including technical inspection weather those drawings are meeting the base line standards and followed by PA, ATP, DTP, CTP and Final approval by CEO. In cases where change in land use permissions have been granted and enforcement actions for unauthorized development and illegal constructions.

Services provided by the urban planning division:

Urban planning team works very effectively in providing services to the citizens and drafting each sector plans and maps which was later processed and digitized by the GIS division.

The basic works done by the urban planning division are:

- Preparing, reviewing and monitoring the implementation of development plans and Formulate development policy, acts, rules, manuals and guidelines for development control, land use and conservation.
- Preparing and coordinating action area plans and urban conservation plans and undertake research (data collection and analysis).
- To disseminate data and information to staff, government agencies and the public on matters related development control and land use planning.

Gaps and challenges:

Apart from the CLU processes the division lagging in research-oriented solutions because when it comes to social infrastructure planning it need a dedicated team to work on.

Before planning any project, it is very necessary to work a little on project-oriented research for designing the methodology process so that a road map is prepared which is lagging in the division.

Due to lot of planning projects which are going parallel it is difficult to get effective recommendations like in infra projects.

Challenges like its little difficult to understand the vast amount of sectorial data produced by HUDA and transferred to the GMDA.

No standard format is finalized and this results in segregating data and converting into excel data sheets and take a lot of man hour time.

Suggestions:

Instead of following conventional process highly dependent on secondary data for planning it is very necessary to adopt the new techniques which are cost effective.

A separate crew should be hired for the primary data collection in digital way so that it’s not necessary to reenter the data in excel sheets which saves a lot of time.

Additional workstations should to add to Interns with internet facilities so that interns can perform the research work faster.

Chapter 3

INTRODUCTION TO HEALTH INFRASTRUCTURE

Every country has its own health care delivery system, but the most common element is primary health care. Health care sector is one of the largest service sectors in India. Its mandate that every state put efforts for raising the health standard and standard of living of the population and improvement of public health as its major priority. Access to health care depends on how health care facility is provided at community level. In India, the healthcare sector shows a good improvement, since last few years. This can be understood by health indicators improvement such as infant mortality, maternal mortality, and life expectancy at birth etc. Apart from these improvements, India still faces many issues and gaps in the healthcare delivery mechanism. The Indian health care system comprises of public health care system in hierarchy private owned hospitals, health personnel, medical colleges, program manager, etc.

There is a considerable shortage of hospitals, hospital beds, and trained medical staff such as doctors and nurses, and so the accessibility among the public is not so good. The rural-urban imbalance also hampers access to health care services. The public health care sector is very poor and responsible for such health status of Indians. The private healthcare sector is mainly responsible for the majority of healthcare in our country. According to NFHS-3, the private healthcare sector still remains the primary source of healthcare for almost 70 per cent of urban households and 63 percent of rural households.

India level Healthcare Initiatives

India's Ministry of Health was established with independence from Britain in 1947. The government has made health a priority in its series of five-year plans, each of which determines state spending priorities for the coming five years. The National Health Policy was endorsed by Parliament in 1983. The policy aimed at universal health care coverage by 2000, and the program was updated in 2002. In 2018 Government of India taken the initiative and the scheme, known officially as Ayushman Bharat or the National Health Protection Mission but more popularly as Modicare, will cover 1300 illnesses, including cancer and heart disease. It is an enormous undertaking and a huge financial commitment, particularly for a country that, to date, has a poor record for health spending. The health care system in India is primarily administered by the states and mandatory to provide quality of health care services for targeted population.

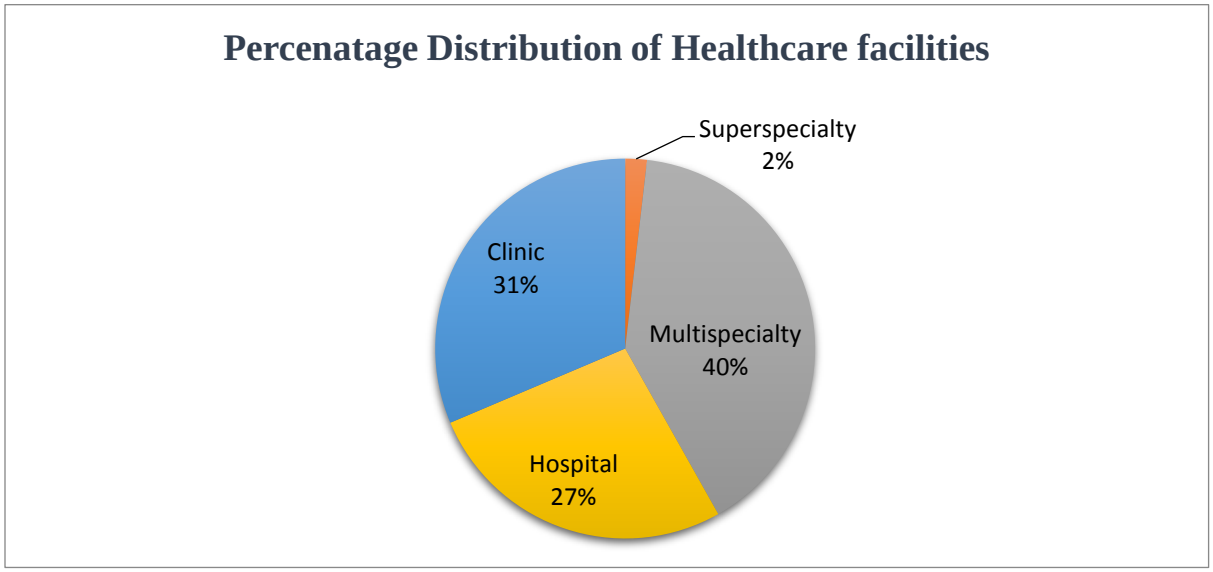
Existing private health infrastructure:

Over the last few years, Gurugram as a city has travelled a long distance and today it has attained a singular status called the millennium city of India for its commercial base. The city is one among the most preferred destination for information technology giants and manufacturing facilities from all over the world. Gurugram will outdo other cosmopolitan cities of India in terms of health care. As a matter of fact, Gurugram is expected to proceed further and to become global medical tourism destinations in the near future. Gurugram meets patients not just from India but from all over the world for its world class super- specialty hospitals equipped with finest doctors in the country and also most preferred health care destination for advanced treatments and offers the best possible medical aid for knee replacements, liver transplants, and kidney transplants, advanced neurosurgeries and so on.

Table 1: Existing private Health Infrastructure Facilities in Gurugram

Type of Service	No. of Facilities	No. of Beds	Ambulances
Super Specialty	6	1950	23
Multispecialty	130	5802	110
Hospital	87	830	17
Clinic	102	379	14
Total	325	8961	164

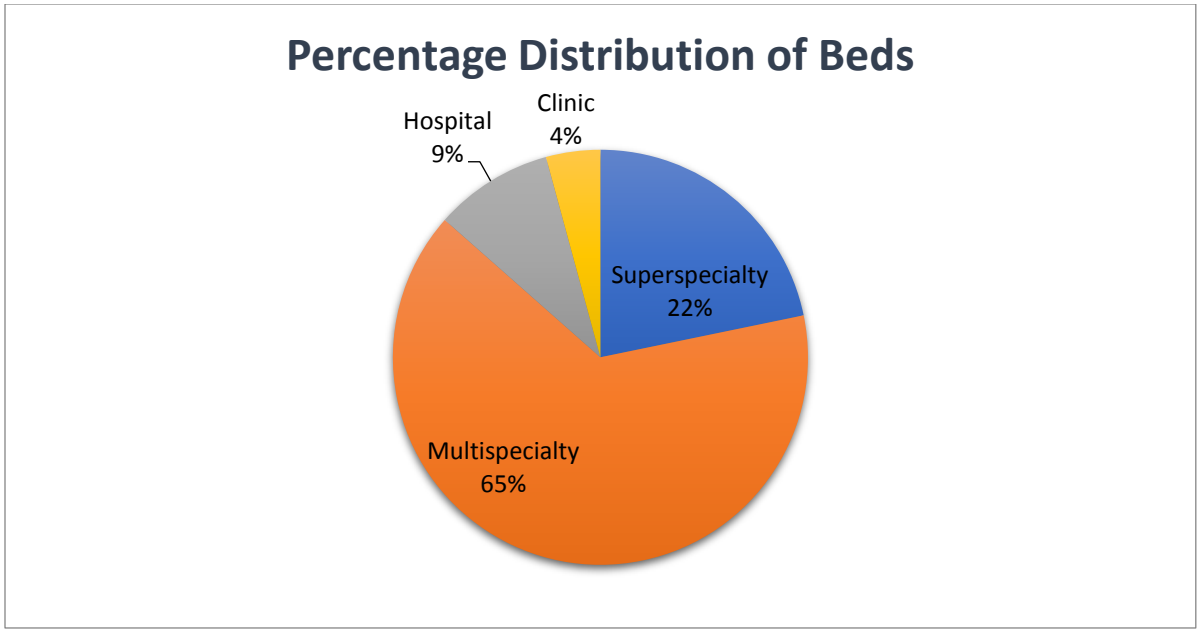
Source: Extracted by Author from GMDA Data, 2019



Source: Generated by Author from GMDA data, 2019.

Figure 3: Percentage Distribution of HealthCare Facilities

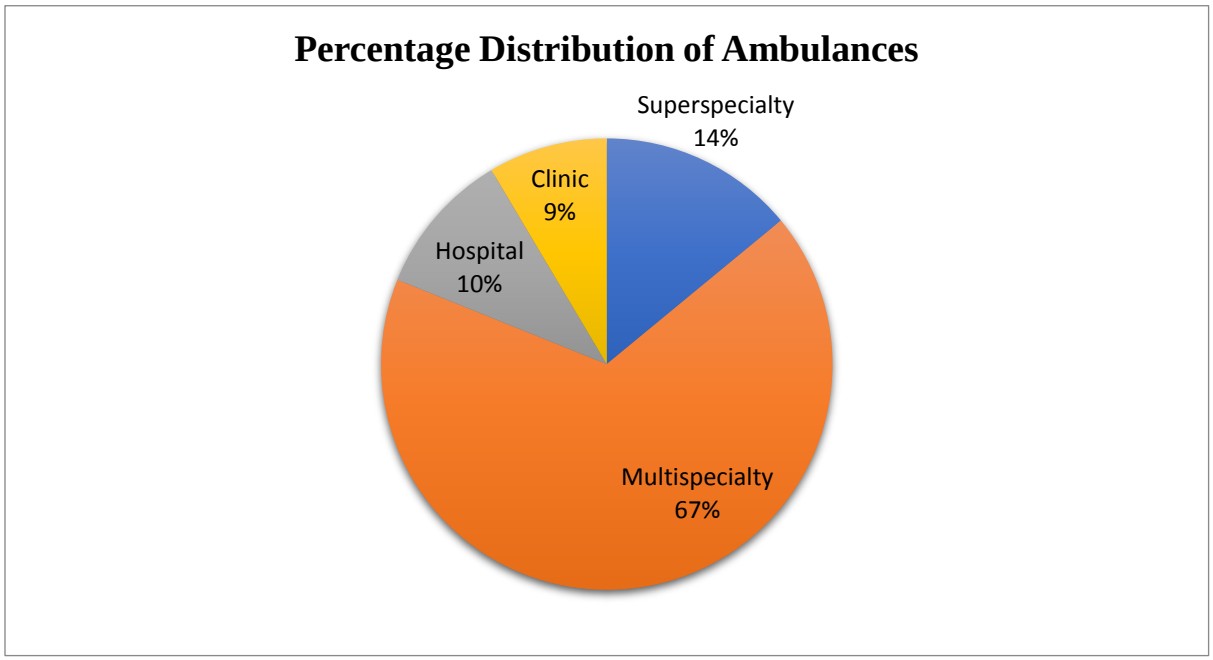
In the above diagram the majority of health services are catered by Multispecialty 40%, Clinics 31%, Hospitals 27% and Super-Specialty by 2%.



Source: Generated by Author from GMDA data, 2019.

Figure 4: Percentage Distribution of Beds

In the above diagram the majority of Number of Bed facilities are catered by Multispecialty 65%, Super-Specialty 22%, Hospitals 9% and Clinics 4%.



Source: Generated by Author from GMDA data, 2019.

Figure 5: Percentage Distribution of Ambulances

In the above diagram the majority of Ambulance services are catered by Multispecialty 67%, Super-Specialty 14%, Hospitals 10% and Clinics 9%.

Various dimensions like affordability and accessibility to Healthcare infrastructure to every individual is needs to be provided for the holistic development of the healthy communities. Therefore, information for all types of infrastructure and their importance needs to be studied at sector level.

Health infrastructure -standard guidelines:

As per the guidelines of URDPFI, HSVP, TCP Haryana, the following norm needs to be followed:

Table 2: URDPFI Guidelines

Sn. No.	Category	Population Served per Unit	Area Requirement in Acre
1	Dispensary	15000	0.2964
2	Nursing home, Child welfare	45000 to 1 Lakh	0.741
3	Polyclinic	1 Lakh	0.741
4	Intermediate Hospital (Category B)	1 Lakh	2.47
5	Intermediate Hospital (Category A)	1 Lakh	9.139
6	Multi-Specialty Hospital (NBC)	1 Lakh	22.23
7	Specialty Hospital (NBC)	1 Lakh	9.139
8	General Hospital (NBC)	2.5 Lakh	14.82
9	Family Hospital Centre	50,000	0.197
10	Diagnostic Centre	50,000	0.197
11	Veterinary Hospital for pets and animal	5 Lakh	0.494
12	Dispensary for pet animals and birds	1 Lakh	0.074

Source: Urban and Regional Development Plans Formulation and Implementation (URDPFI) Guidelines, Government of India

Table 3: HSVP Guideline

S. No.	Category	Land Required in Acres	Remarks
1	Dispensary	0.25	**Internal Community Building Norms** Land for each sector. one built up dispensary for every two sectors.
2	50 Bedded Hospital for 1 lacs Population.	1.00	**External Community Buildings Norms** 50 Bedded Hospital for 1 lacs Population.
3	One No. 500 bedded Hospital for towns having Population more than 10 lacs.	10.00	**External Community Buildings Norms** One No. 500 bedded Hospital for towns having Population more than 5 lacs.

Source: HSVP

Table 4: TCP Haryana Guideline

S. No.	Category	Land Required in Acres	Remarks
1	Hosital	• 1 Acre • 10.00 Acres	• External community building norms:- • 50 bedded hospital for 1.00 lacs population. • one no. 500 bedded hospital for towns having population more than 5 lacs.
2	ESI Dispensary	1.0 Acre	• External community building:- • for every 10,00 to 15,00 population (industrial labor) in industrial sectors.
3	Dispensary/ Health Facility Centre	1.25 Acre	• Area & population based for norms provision of internal community buildings in licenced colonies • one for every 10,000 to 15,000 population
4	Health Centre/ Health Facility	2.0 Acre	• Area & population based for norms provision of internal community buildings in licenced colonies • one for every 30,000 population
5	Nursing Home		• Area & population based for norms provision of internal community buildings in licenced colonies • 2 no's 1000 sqm. for each 50.00 acres.
6	Clinic		• Area & population based for norms provision of internal community buildings in licenced colonies • 2 no's 250 sqm. for each 50.00 acres

Source: TCP Haryana.

Chapter 4

REVIEW OF LITERATURE:

India is 2nd In the last few years Indian cities are considering social infrastructure is the one of the major components for economic growth and also improves the neighborhood relations of the citizens. In the mean while private health infrastructure continually increasing in terms of delivery of healthcare facilities (Kumar & Gupta, 2012) author also suggested to introduce the geo coding and geographic information system for continuous monitoring of disease and service delivery.

Geographical Information Systems is an integrated tool which is important for the analysis and the decision making which includes land, environment and humans A complete definition has been given by the International Federation of Surveyors, (FIG – 1983) according to w hich: 'A Geographical Information System is a system for the decision process of legal, admi nistrative, and financial decision for designing, enhancing of a Data Base for specific regions and sector specific also. The base of a GIS is a unified geographical system which also makes data linking more convenient between them, as well as with other system including data with reference to land. Spatial Analysis on the other side, is focused on developing methods for the evaluation of existing and proposed characteristics of spatial organization, and can be defined as 'a series of methods and tools aimed to study and analyze geographical information' (Huxhold, 1991).

The author collected the data of existing health infrastructure in the metropolis using GPS and the date and is converted into geospatial data base and analyzed in the ArcGIS software. By processing the data identified the percentage share of public and private health infrastructure and also used buffer to analyze with a 1.5km radius and identified the lacking of health facilities in the areas that doesn't spread some parts of the city area. The author concluded with pointing the importance of primary health centers which enhances the standard of living socially and economically by providing 24 hours service in major centers. Disease based management system should be established with a special GIS based team to take effective decision making by the state health ministry. Author also notified capacity building programs to nurses and paramedical staff plays a crucial role in delivering the quality of health services and strategic planning for health and disease prevention education on some notified areas in the metropolis.

In this paper the author discussing the advantage of using spatial analysis for health sciences in planning, mapping, locating the existing health infrastructure, integrated dynamic maps facilities and importance of inclusive data collection and maintenance of the data through central servers. In conventional methods the data collection and processing are cost intensive and no idea of dynamic maps. But in GIS by using different tools like overlay analysis which helps in making hypothesis over different layers and utilized for critical spatial decisions in healthcare sector. Other analysis tools like buffer analysis, network analysis, statistical analysis, web GIS and extrapolation. By using these techniques, we can produce the data integrated infographic maps like analyzing the spread of diseases to what extent in a particular area and take decisions at various levels. Accessibility to healthcare facilities in emergency situations to the citizens as well as ambulances is very difficult because less awareness on nearest and good connectivity of transport lines which can be sorted out in GIS by integrating it. In conclusion by utilizing GIS Based health infrastructure mapping can easily asses the existing data and population density norms.

OBJECTIVES

Aim: To prepare a GIS based Strategic frame work for Gurugram metropolitan area on Private Health Infrastructure at Sub city level.

Objectives:

1. Map and Analyzing the buffer zones to existing private health Infrastructure at city level with existing GIS data.
2. To propose a frame work for healthcare infrastructure at sub city level according to population projection for 2031.

Research Questions:

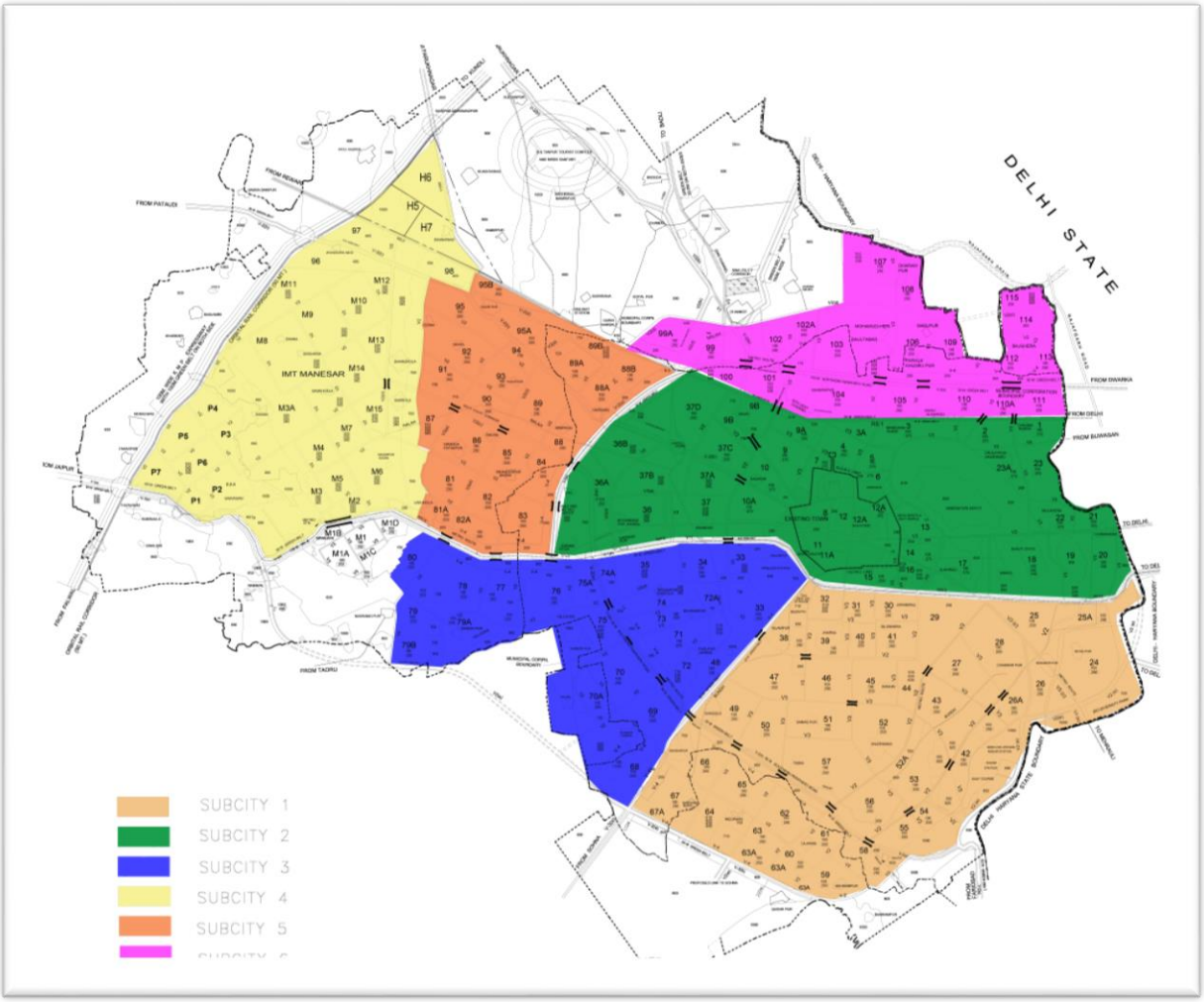
1. What is the need for Decentralized private Health Infrastructure for a metropolitan city like Gurugram?
2. How integrated Health care facilities can help in delivering the quality of service to the citizens?

METHODOLOGY:

Literature Review	• Collecting data from journals,reports and news for understanding the importance of the health infrastructure palnning.
Secondary Data	• Data will be gathered through urban planning and GIS division and also from other organizions like census of India, HUDA, Haryana health portal and Ministry of Health and Family Welfare.
Data Analysis	• Data is segregated and analyzed and converting the data into pictorial representationand also processed in GIS basesed Analysis tool.
Gap Identification	• Identifing the gaps in existing healthcare infrastructure at subcity level.
Strategic planning	• Preparing a strategic frame work for planning according to URDPFI guide line.
Recomendations	• According to the outputs the recomendations should be proposed.
Conclusion	• Concluding the project as per the terms

INTRODUCTION TO STUDY AREA:

Gurugram city located in Haryana and is near the National Capital Region of India and 268 kilometers southwest of Chandigarh, the state capital and 32 kilometers southwest of New Delhi. According to 2011 census, Gurugram had a population of 15, 14,432.



Source: Urban Planning Division, GMDA

Figure6: Sub-city Map of Gurugram

The total area of Gurugram district is 732 square kilometers and GMDA area is 675 sq.km and Gurugram Manesar Urban Complex(GMUC) area is 462.15 sq.km in which GMUC is divided into 6 sub cities and each subsidy is divided into sector roads basis on national highway roads and state road crossing through the city also on growth patterns like old and new Gurugram.

Table 5: sub-city wise area and Population including TOD Projections

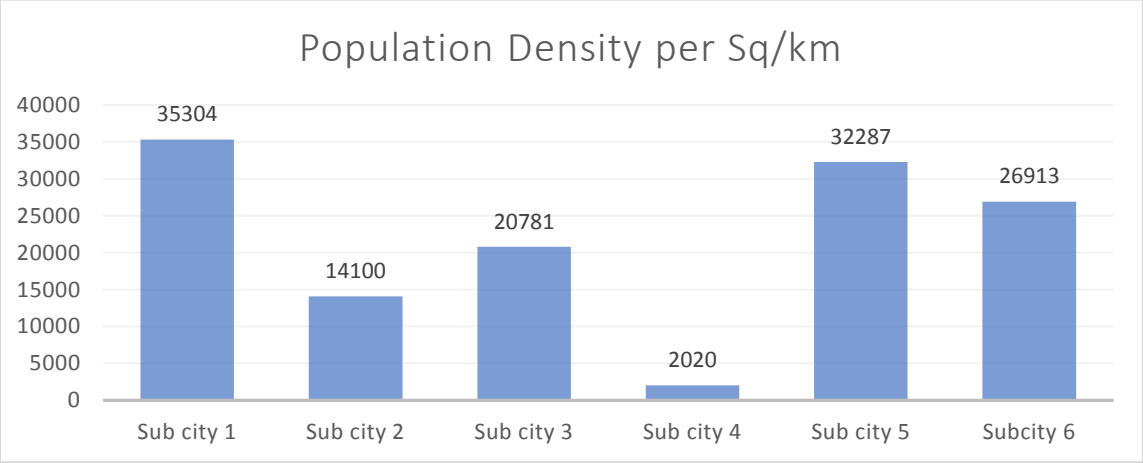
Sub city		Area in Sq.kms		Population	Villages		No. of sectors	Population Density per Sq/km
Sub city-1		68.6		2428964	18		46	35304
Sub city-2		80.4		1133653	11		41	14100
Sub city-3		46.2		960121	7		23	20781
Sub city-4		57.7		116606	10		33	2020
Sub city-5		35.8		1155900	10		23	32287
Sub city-6		34.4		925821	8		20	26913

Source: Extracted by Author from GMDA data, 2019.

World is urbanizing at a faster rate towns, cities and Urban agglomerations are acting as potential economic drivers for developing countries in South Asian continents. All though cities are experiencing a shift in economic activities and in the last two decades India entered into new phase of Urbanism due to Advancement in Information technology, Availability of cheap human capital and natural resources like land and it contributed GDP from 1.1 in 1998 to 7.7 in 2017. Metropolitan cities like Delhi-NCR, Mumbai, Pune, Chennai, Kolkata, Bangalore, Hyderabad and new emerging cities like Visakhapatnam acting as IT-HUBs for global Markets. Over a time period increase in Natural Population growth and demographic transition distributed the population to Peri-Urban Areas. In this case of Gurugram from a small town in the last four decades emerged as the Millennium city of the India's Growth and Rapid Development at the global platform. So, it very necessary to understand the growth journey of Millennium city from ancient times to till day. This transformation can be classified into different phases of the town's growth. The town's growth dynamics includes both the spatial dynamics and population dynamics experience and

involved multiple players such as the government, the residents and the private developers. In its growth journey, the city space has got divided and changed into differential spaces coexisting within the city.

Due to rapid urbanization, Gurugram has become a leading Financial and Industrial hub with the third highest per capita income in India. The city's economic growth gain momentum when the leading Indian automobile manufacturer Maruti Suzuki India Limited established a manufacturing plant in Gurugram in the 1970s. In present situation Gurugram has local offices for more than 250 Fortune 500 companies. Gurugram from a small town in the last four decades emerged as the Millennium city of the India's Growth and Rapid Development at the global platform. A Very Few cities in India experienced the growth at this pace. With all its aspirations and dreams Gurugram emerging as global city in India with a little time period city transformed and become an economic and development Hub providing the center for rapid growth in all fields and services at international platform. At the same time rapid development has created its own set of issues and challenges at city level like poor mobility and uncontrolled population density growth. New legal structures that provide sustainable solution for future growth with challenges are being proposed for inclusive development.



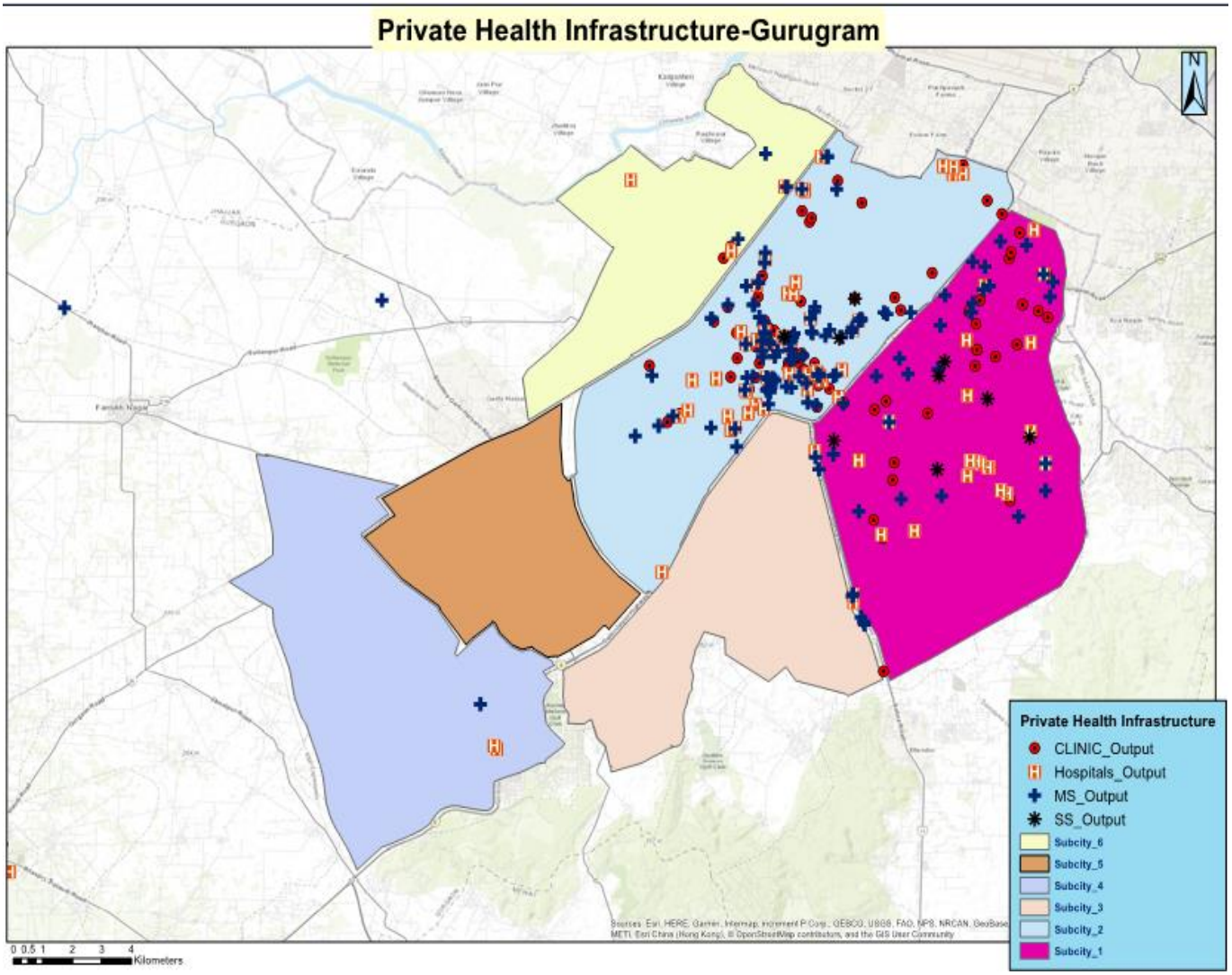
Source: Generated by Author from GMDA data, 2019.

Figure 7: Population density of sub city

Chapter 5

BUFFER -RADIAL ANALYSAS.

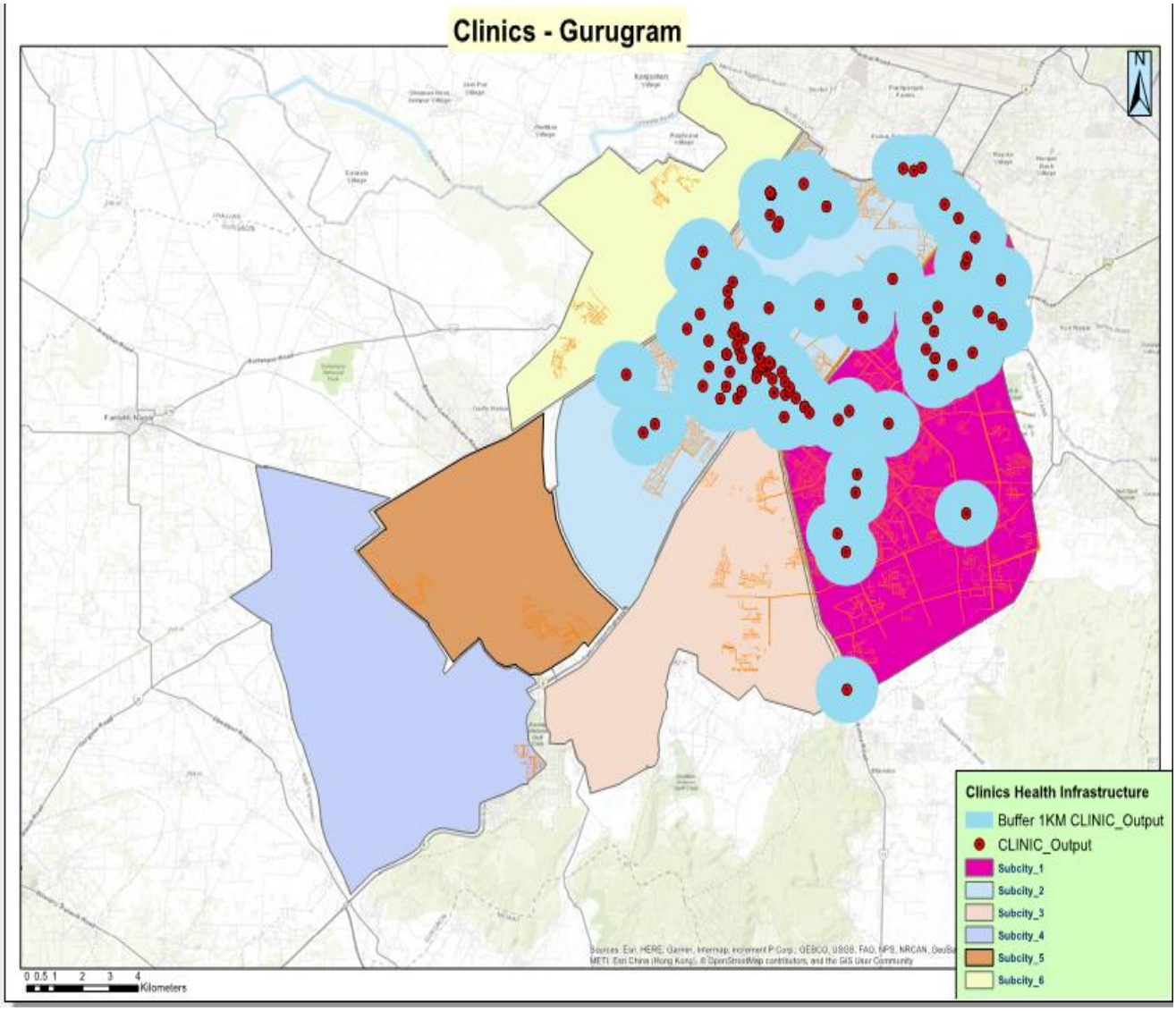
The latitude longitude data gathered from GIS division is processed in ARCMAP software and located the existing health infrastructure in and around the city and then performed buffer analysis tool. In the below map by spatial mapping it is showing that most of the private health infrastructure consists of clinics, hospitals are concentrated in sub city 1 and sub city 2 and sub cities 3,5 didn't have any facilities and sub city 4 and 6 have partial health infrastructure.



Source: Generated by author in Arcmap,2019

Figure 8: Private Health Infrastructure-Gurugram.

Clinic Infrastructure Buffer Analysis:

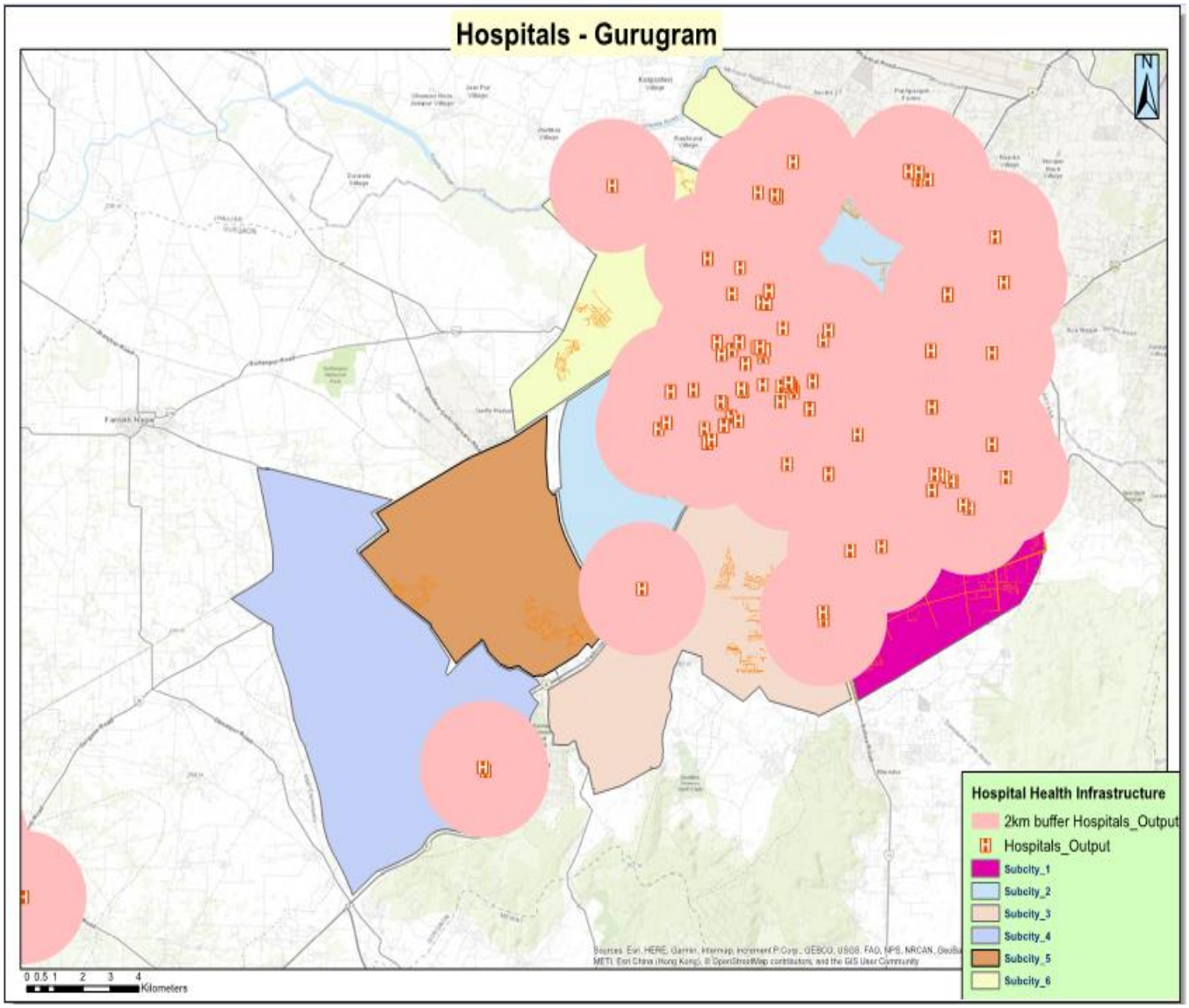


Source: Generated by author in Arcmap,2019

Figure 9: Clinics buffer zone of 1 km

In this map a buffer zone is created with 1 km buffer radius because according to URDPFI guidelines any facility must be available with in 800 meters as Gurugram is well connected with sector roads. If we observe that clinical services majorly concentrated in sub city 2 and scattered way in sub city 1 and rest of 4 sub cities didn't have any clinical services. By this while planning or approving for private clinics priority should be given to the sub cities which are lagging in infrastructure.

Hospital Infrastructure Buffer Analysis:

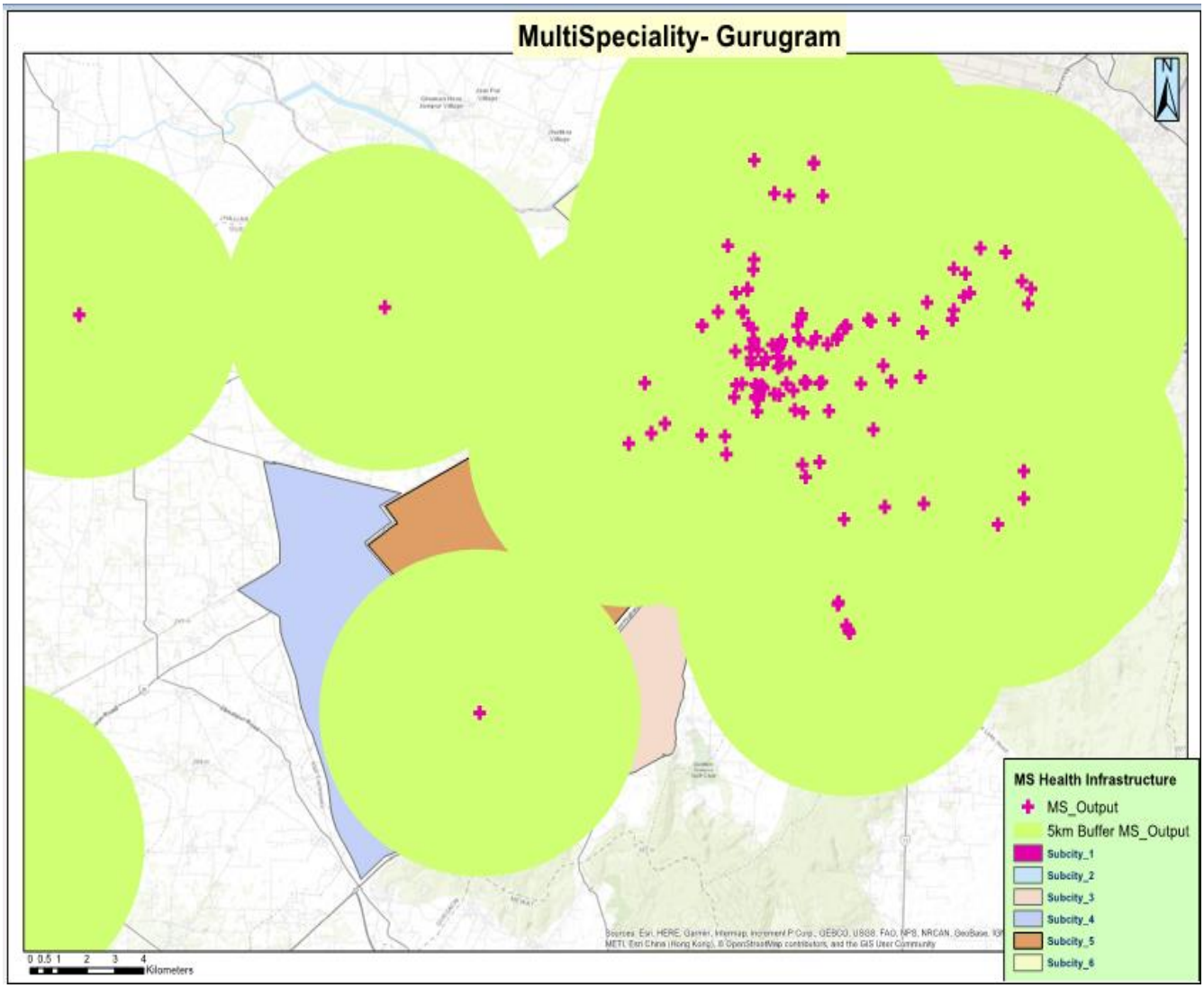


Source: Generated by author in Arcmap,2019

Figure 10: Hospitals buffer zone of 2 km

In the above hospitals date are segregated and a buffer radius of 2 km is taken. By the map with buffer zone we can easily analyze that sub cities sub cities 3,4 and 5 are lagging with private health infrastructure and partially sub city 1,2 and 6. So for planning for private hospitals approvals must be in priority basis for lagging areas in sub cities.

Multi-Specialty Infrastructure Buffer Analysis:

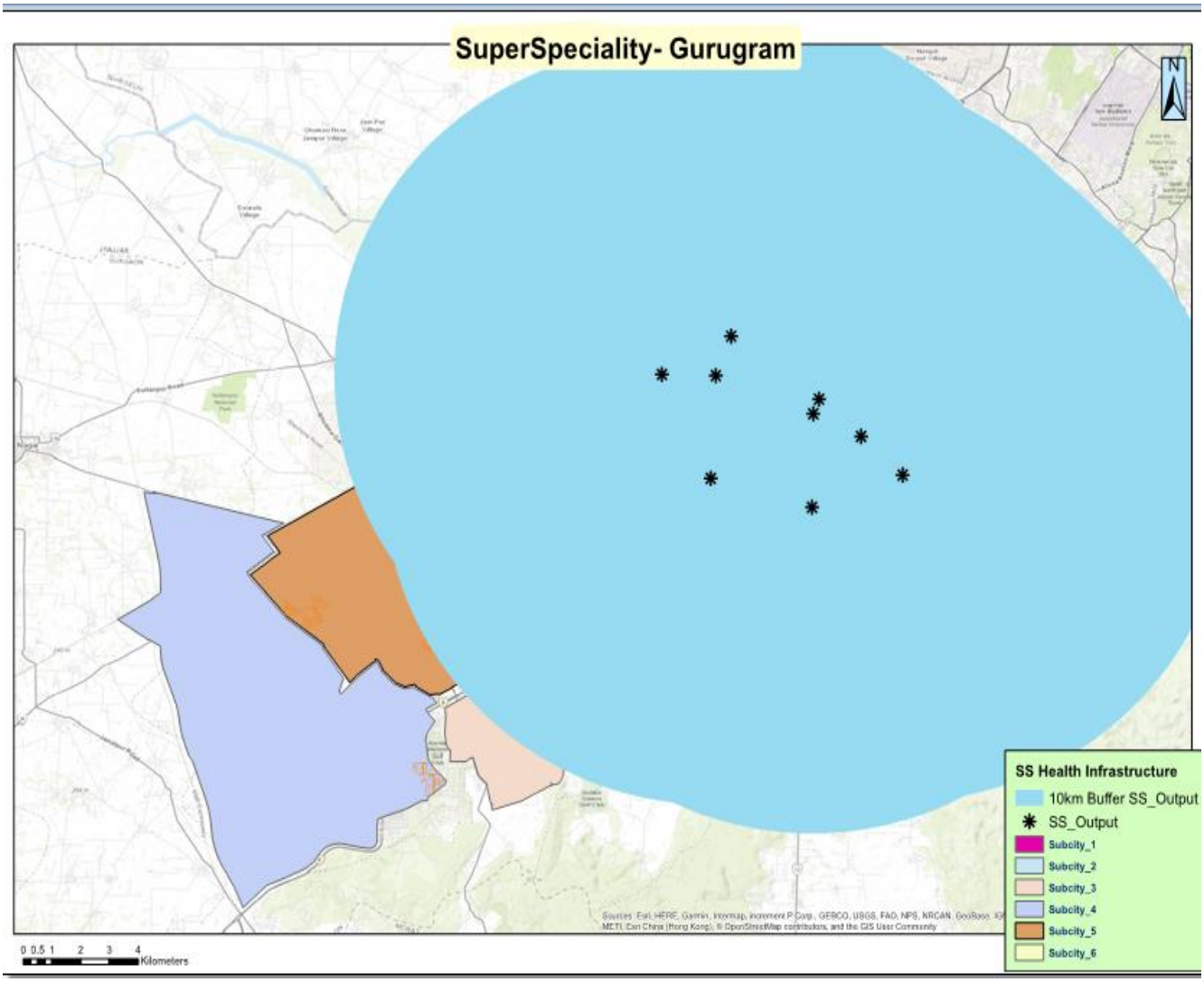


Source: Generated by author in Arcmap,2019

Figure 11: Multispecialty buffer zone of 5 km

In the above map for Multi-specialty a 5km buffer radius is considers because this type of infrastructure is mainly used on emergency and special treatments.by this analysis we can easily notice that majority of the private infrastructure is located in sub city 2 and scattered way in sub city 1. With 5 km buffer almost GMUC is covered but still some parts in sub city 3,4 and 5 still not in the proximity. So for the future planning these areas are considered in priority.

Super-Specialty Infrastructure Buffer Analysis:

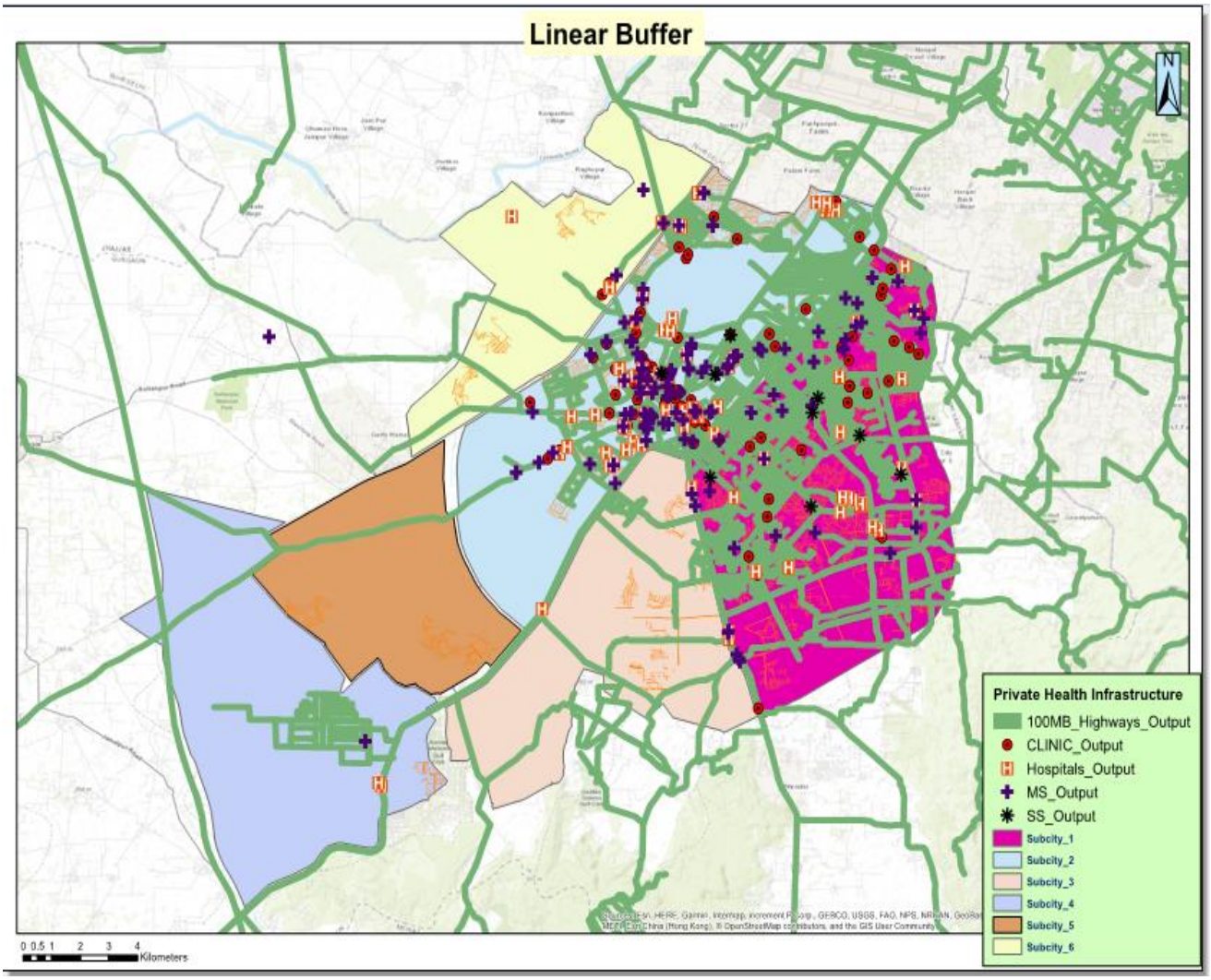


Source: Generated by author in Arcmap,2019

Figure 12: Super specialty buffer zone of 10 km

In the above map for Super-specialty a 10km buffer radius is considers because this type of infrastructure is mainly used on emergency and major treatments.by this analysis we can easily notice that majority of the private infrastructure is located in sub city1 and sub city2.With 10 km buffer almost GMUC is covered but still some parts in sub city 3,4 and 5 still not in the proximity. So for the future planning these areas are considered in priority. Population density is very important parameter while planning according to the norms.

Linear Buffer Analysis-Private health Infrastructure:



Source: Generated by author in Arcmap,2019

Figure 13: Linear buffer analysis 100 meters

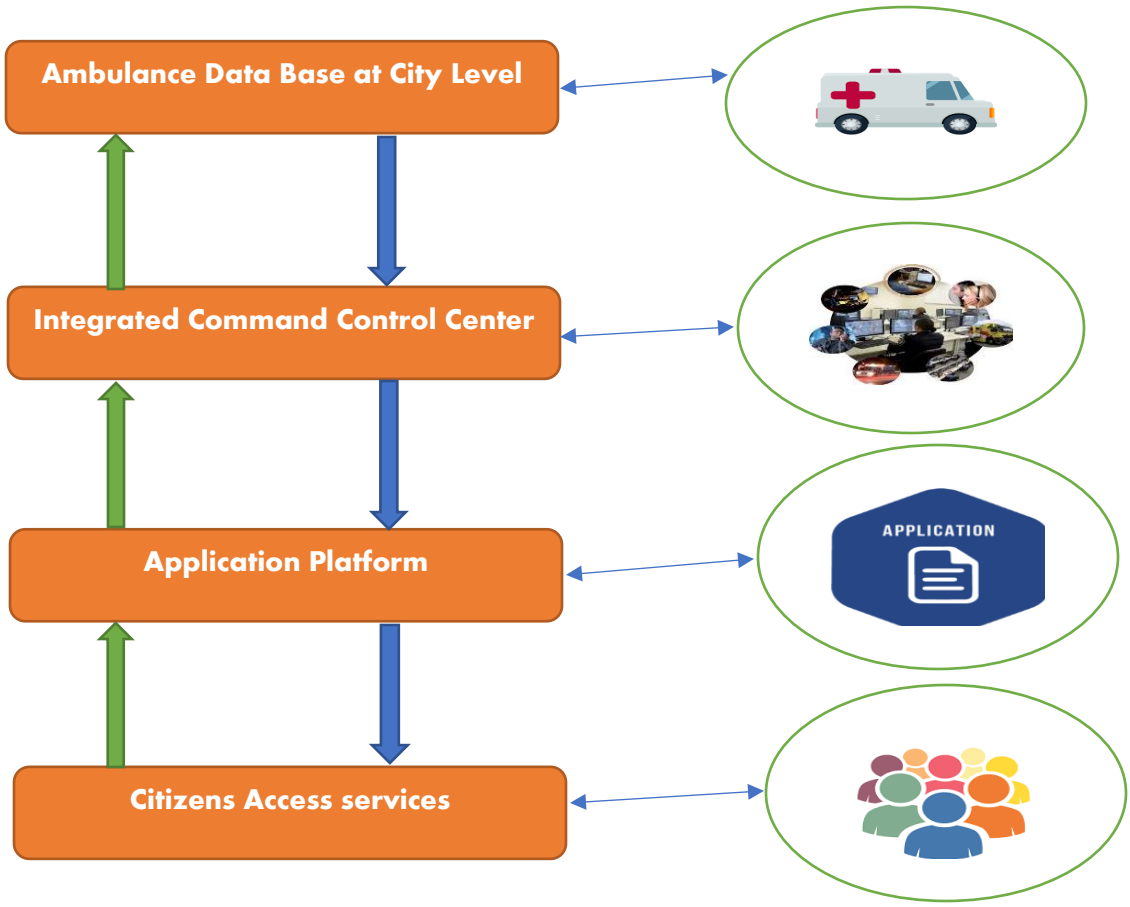
In the Linear buffer analysis, a 100 meters buffer zone is generated on major road networks like national highways, state highways and sector roads also included in sub city 1 and sub city 2 to analyze the accessibility to the Private health infrastructure. By the above map its is analyzed that maximum number of Private health care facilities are connected and accessible to commutate.

Innovative Suggestion: Integrated Ambulance Service Management

Gurugram is experiencing a rapid development and population also increasing at a faster rate. To cater Ambulance services by individual players it is difficulty and as of now there is no such platform for citizens as well as for migrant population to access the city level ambulance facility in Gurugram. In view of sustainable development ambulances are the city assets and planned utility improves the quality of service provided to the citizens as well as increases the efficient usage of resources. As Gurugram Metropolitan development authority is planning to develop a integrated single platform application for citizens. It is suggested to add this **IAFM** platform in the application which will help citizens to access the Ambulances services of any emergency situations. Majority of the ambulance assets are with multispecialty followed by superseniority.

If this service is integrated with citizens application may be one of Innovative solution to utilize the city resources efficiently as Gurugram is stepping towards smart city as well as in competitive cities on global platforms. This innovative solution may stand an example for other developing cities and urban agglomerations in India.

Working Module:



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Learning outcomes:

- Day to day working with government body.
- Learnt basics of planning and methods.
- Learnt data extraction from different sources and data analytics.
- Gained real time knowledge and importance spatial analysis is planning.
- Understand the complexity of urban planning.

Conclusion:

- Infrastructure mapping is very important in now a day especially when it comes to health infrastructure.
- During the internship period the study conducted on private health infrastructure for the city Gurugram.
- Particularly Gurugram Manesar Urban Complex (GMUC) with the available latitude and longitude data sets and extracted.
- Analyzed through various tools like linear buffer and radius buffer and mapping the data sets.
- If these data sets include public hospitals then we can find the gaps in health infrastructure with population density at sector level according to the guidelines and norms.

References

1. Aliyu, Y. and Shebe, M. (2013). Using GIS in the Management of Health Infrastructure within Kaduna Metropolis, Nigeria. *Mediterranean Journal of Social Sciences*.

2. Applications of GIS in Health Sciences. (2009). *Shiraz E-Medical Journal*, [online] 10(4). Available at: http://semj.sums.ac.ir/vol10/oct2009/87055.htm [Accessed 10 Aug. 2019].

3. Xhafa, S. and Kosovrasti, A. (2015). Geographic Information Systems (GIS) in Urban Planning. *European Journal of Interdisciplinary Studies*, 1(1), p.85.

4. Barton, H. and Grant, M. (2012). Urban Planning for Healthy Cities. *Journal of Urban Health*, 90(S1), pp.129-141.

5. Haryanahealth.nic.in. (2019). *Health Department, Haryana*. [online] Available at: http://haryanahealth.nic.in/ [Accessed 18 Jul. 2019].

6. Haryana, E. (2019). *Economic Survey of Haryana*. [online] Esaharyana.gov.in. Available at: http://esaharyana.gov.in/en-us/Economic-Survey-of-Haryana [Accessed 18 Jul. 2019].

7. Mohfw.gov.in. (2019). *Home | Ministry of Health and Family Welfare | GOI*. [online] Available at: https://mohfw.gov.in/ [Accessed 18 Jul. 2019].

8. Ncrpb.nic.in.(2019).[online]Availableat:http://ncrpb.nic.in/pdf_files/16ch12social infrastructure.pdf [Accessed 19 Jun. 2019].

9. Study on Health Infrastructure in NCR. (2015). [online] New Delhi: National Capital Region Planning Board. Available at: http://ncrpb.nic.in/pdf_files/NCRPB%20Health%20Final%20Report%20Vol%20I-Final_Feb%202016.pdf [Accessed 18 Jul. 2019].

10. Smart Healthcare System. (2017). *International Journal of Recent Trends in Engineering and Research*, 3(3), pp.198-200.

11. Lefebvre, B. and Chapelet, P. (2019). *Contextualizing the Urban Healthcare System. Methodology for developing a geodatabase of Delhi's healthcare system*. [online] Hal.archives-ouvertes.fr. Available at: https://hal.archives-ouvertes.fr/hal-00686947 [Accessed 29 Aug. 2019].

12. An Outlook on India's Healthcare System with a Medical Case Study and Review on Big Data and its Importance in Healthcare. (2016). *International Journal of Science and Research (IJSR)*, 5(4), pp.1486-1492.

13. Rekha, R., Wajid, S., Radhakrishnan, N. and Mathew, S. (2017). Accessibility Analysis of Health care facility using Geospatial Techniques. *Transportation Research Procedia*, 27, pp.1163-1170.